

Ako Baloyi

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Education

University of the Witwatersrand Johannesburg, South Africa
Bachelor of Engineering Science (Digital Arts) Expected Nov 2026
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Software Engineering, Embedded Systems

Giyani High School Giyani, South Africa
National Senior Certificate (Matric) Nov 2022

Technical Skills

Programming: Python, Java, C++, C#, AVR Assembly, JavaScript

Web Technologies: Node.js, Express, HTML, CSS

Cloud & Infrastructure: AWS (Amazon S3, AWS Amplify)

Tools: Git, Android Studio

Domains: Embedded Systems, Cybersecurity, Systems Architecture, Cloud Integration

Projects

Web-Based Morabaraba Game Platform JavaScript, Node.js, Express, HTML, CSS
Live Demo — GitHub Repository

- Engineered a full-stack web implementation of the traditional Morabaraba board game using Node.js and Express.
- Designed a deterministic game engine responsible for board state management, move validation, mill detection, and capture logic.
- Built a browser-based interface using HTML, CSS, and JavaScript that synchronizes player actions with the underlying game logic.
- Structured the application with clear separation between frontend rendering, user interaction, and server-side game rules.

Interrupt-Driven Safe Lock System ATmega328P, AVR Assembly

- Developed a keypad-based electronic safe lock system programmed entirely in AVR assembly.
- Implemented interrupt-driven architecture using Timer1 PWM and pin-change interrupts for responsive input handling.
- Integrated matrix keypad scanning, I2C LCD display via PCF8574, and servo motor control for the locking mechanism.
- Designed a non-blocking state machine for password entry, validation, LCD feedback, and lock actuation without delay loops.

SecureCloud Android Vault

Java, Android Studio, AWS

- Built a proof-of-concept Android application for secure file storage and retrieval using AWS cloud infrastructure.
- Implemented Amazon S3 and AWS Amplify for backend storage, authentication, and client connectivity.
- Connected the Android client to cloud services through REST-based request flows for uploading and retrieving user data.

Certifications

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| • Fortinet Certified Associate Cybersecurity — Fortinet | Nov 2025 |
| • Fortinet Certified Fundamentals Cybersecurity — Fortinet | Nov 2025 |
| • Fortinet FortiGate 7.6 Operator — Fortinet | Nov 2025 |
| • Technical Introduction to Cybersecurity 3.0 — Fortinet | Nov 2025 |
| • Getting Started in Cybersecurity 3.0 — Fortinet | Nov 2025 |
| • Career Essentials in Cybersecurity — Microsoft | Jun 2024 |
| • AWS Cloud Quest: Cloud Practitioner — Amazon Web Services | May 2024 |
| • Fundamentals of Digital Marketing Certification — Google | Dec 2021 |